Reading group notes: Kolmogorov-Arnold networks

Reading group of the Ashbury machine learning lab

1 Background

Recent work has made the residual activation a central concern, because it decides how far Kolmogorov-Arnold network models can scale. We argue that the layer width deserves as much attention as the symbolic formula. This work studies Kolmogorov-Arnold networks, a method that rethinks the spline grid. Most prior methods treat the symbolic formula as fixed and tune the grid interval around it.

Combining Kolmogorov-Arnold networks with methods that target the pruning threshold is a promising direction. We expect the spline grid to interact with the grid interval at larger scales, which we leave to future work. Our analysis covers models up to 1.3B parameters, and larger models may behave differently. The pruning threshold also makes Kolmogorov-Arnold networks easier to inspect, since its behaviour can be read off per token.

Practitioners report that the grid interval is the first component to fail on long inputs. This work studies Kolmogorov-Arnold networks, a method that rethinks the residual activation. Earlier Kolmogorov-Arnold network methods improved the collocation point but left the grid interval unchanged. Most prior methods treat the collocation point as fixed and tune the spline grid around it.

2 Methods we read

The layer width in Kolmogorov-Arnold networks is parameterised by a small linear layer with 32 hidden units. Each block of Kolmogorov-Arnold networks applies the layer width after a short convolution of width 7. Gradients through the symbolic formula are clipped at a norm of 5.0. The residual activation is initialised so that Kolmogorov-Arnold networks starts training close to a fixed-grid KAN.

The grid interval in Kolmogorov-Arnold networks is parameterised by a small linear layer with 64 hidden units. To keep training stable, Kolmogorov-Arnold networks normalises the layer width before it is combined with the univariate function. We implement the spline grid as a fused kernel, which avoids writing intermediate results to memory. During inference Kolmogorov-Arnold networks caches the univariate function, so each new token costs a constant amount of work.

The pruning threshold is initialised so that Kolmogorov-Arnold networks starts training close to an MLP. Unlike an MLP, Kolmogorov-Arnold networks updates the edge function only when the pruning threshold changes. During inference Kolmogorov-Arnold networks caches the symbolic formula, so each new token costs a constant amount of work. A coefficient controls how strongly the pruning threshold depends on the univariate function; we set it to 0.5 throughout.

3 Results we discussed

The gains of Kolmogorov-Arnold networks grow with the input length, as the residual activation becomes the bottleneck. We train every model for 70B tokens with the same data order and report the mean of three seeds. With the collocation point frozen, Kolmogorov-Arnold networks still recovers most of its improvement on the Feynman equations. On the Feynman equations, Kolmogorov-Arnold networks improves over a Fourier feature network by 6.4 points.

Ablating the spline grid lowers the a PDE suite score by 3.4 points, the largest drop among all components. Variance across seeds stays below 0.3 points for every configuration we report. Throughput is measured with a batch size of 128 on a single accelerator. With the spline grid frozen, Kolmogorov-Arnold networks still recovers most of its improvement on a PDE suite.

Ablating the collocation point lowers the a PDE suite score by 1.9 points, the largest drop among all components. The gains of Kolmogorov-Arnold networks grow with the input length, as the residual activation becomes the bottleneck. Throughput is measured with a batch size of 64 on a single accelerator. With the residual activation frozen, Kolmogorov-Arnold networks still recovers most of its improvement on a tabular regression suite.

4 Open questions

Combining Kolmogorov-Arnold networks with methods that target the pruning threshold is a promising direction. A limitation of Kolmogorov-Arnold networks is that the grid interval must be tuned separately for each model size. Our analysis covers models up to 1.3B parameters, and larger models may behave differently. The basis function also makes Kolmogorov-Arnold networks easier to inspect, since its behaviour can be read off per token.

Combining Kolmogorov-Arnold networks with methods that target the residual activation is a promising direction. A limitation of Kolmogorov-Arnold networks is that the edge function must be tuned separately for each model size. We expect the basis function to interact with the pruning threshold at larger scales, which we leave to future work. Our analysis covers models up to 1.3B parameters, and larger models may behave differently.